

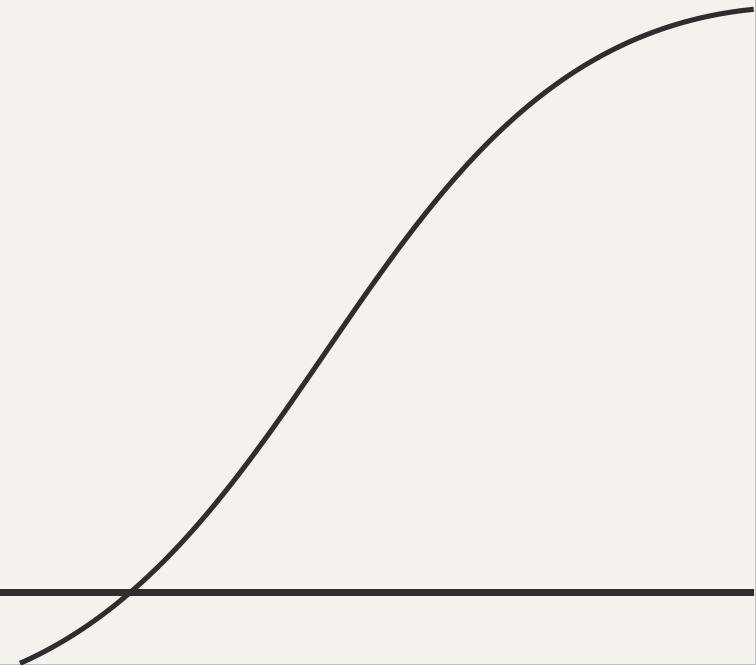
Data Visualization and Storytelling

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March 28, 2026

SESSION 8: The art of storytelling, best practices & delivery



Cycle of Visualization

Chart

Visual representation of information

Focuses on a few variables

Dashboard

A collection of charts that presents information about various variables in one location

Audience Driven and Interactive

Data Story

A narrative that uses charts along with text to convey a clear message or conclusion

Author Driven and static

What is data storytelling?

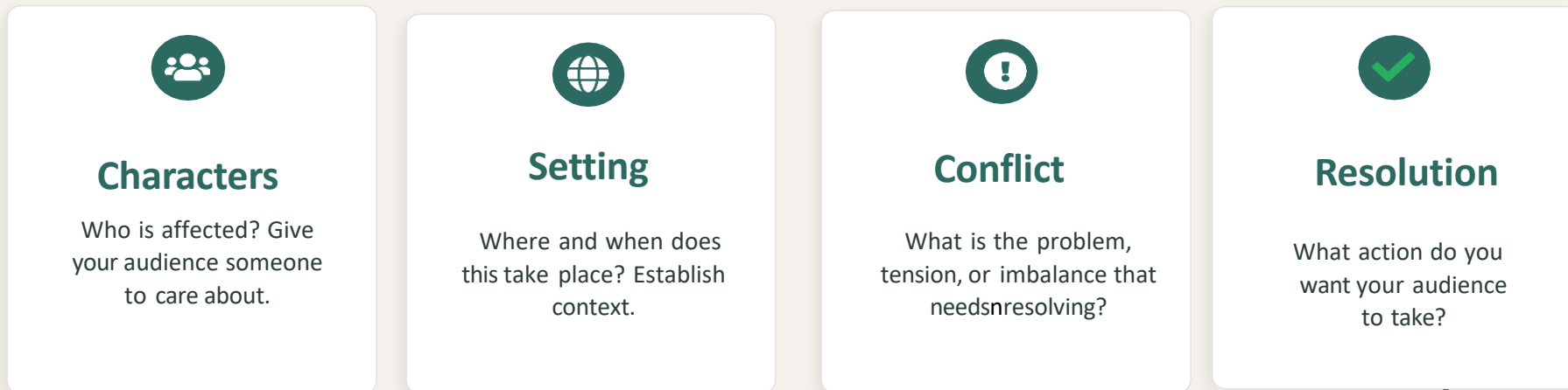
Data storytelling is the ability to effectively communicate insights from a dataset using narratives and visualizations. It can be used to put data insights into context for and inspire action from your audience → **Soft skill**

- I. **Data:** Analyzing data using descriptive, diagnostic, predictive, and prescriptive analysis can enable you to understand its full picture.
- II. **Narrative:** A verbal or written narrative, also called a storyline, is used to communicate insights gleaned from data, the context surrounding it, and actions you recommend and aim to inspire in your audience.
- III. **Visualizations:** Visual representations of your data. These can be charts, graphs, diagrams, pictures, or videos.

How we can use concepts of storytelling to communicate effectively with data?



We can use stories to engage our audience emotionally in a way that goes beyond what facts can do.



How we can use concepts of storytelling to communicate effectively with data?



We can use stories to engage our audience emotionally in a way that goes beyond what facts can do.

Keep it simple.

Edit ruthlessly.

Be authentic.

Don't communicate for yourself—communicate *for your audience*.

The story is not for you; the story is for them.

How to Construct Your Story

1

Introduce the plot

Build context for your audience. Get everyone on common ground. Establish the situation before introducing the problem. Think of this as 'setting the scene.'

2

Develop the conflict & convince

Convince your audience of the need for action. Provide background, external context, comparison points, and data that demonstrates the problem. Articulate what happens if no action is taken.

3

Call to action

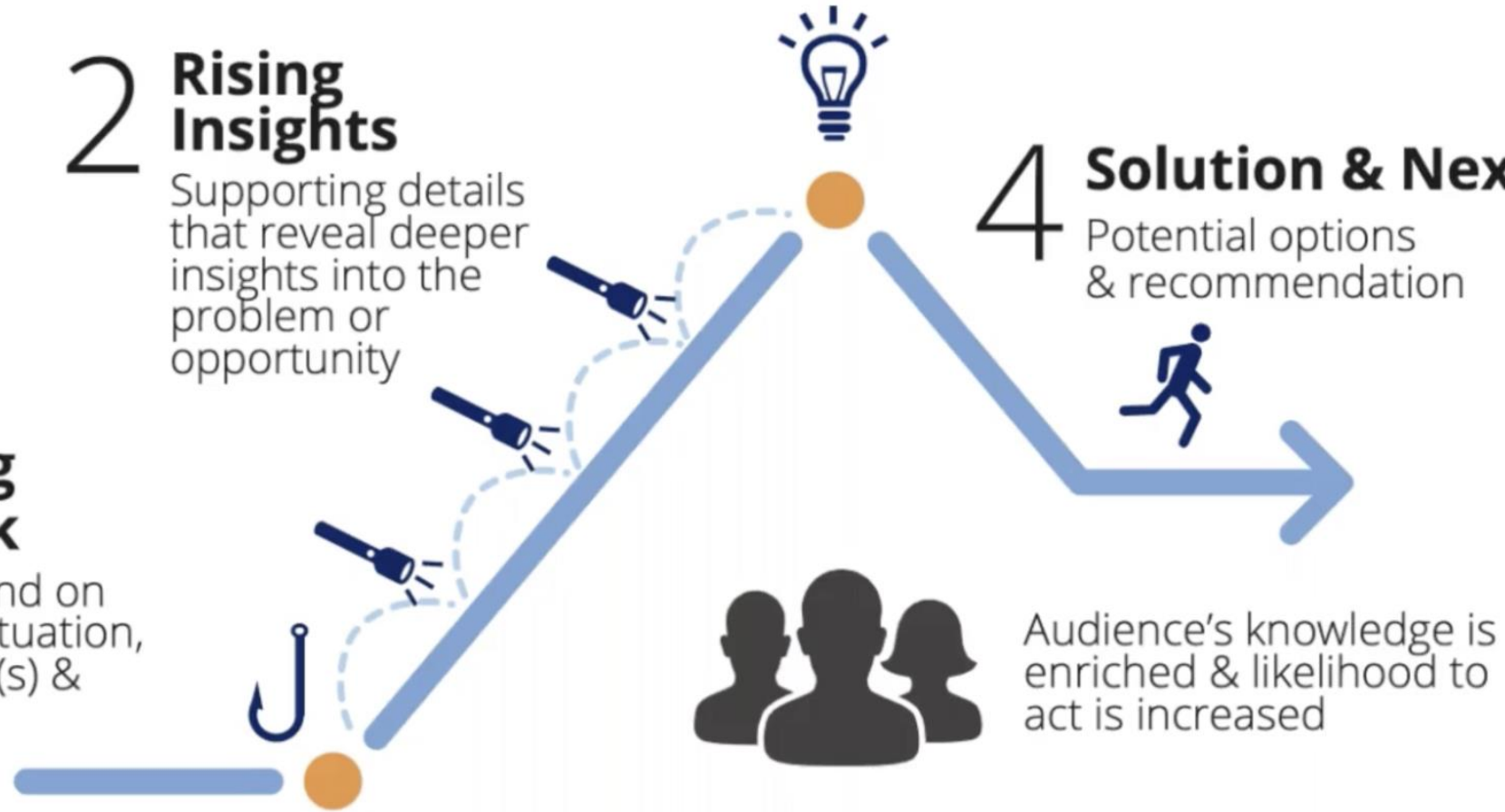
Make it totally clear what you want your audience to do. Recap the problem and urgency. Send them off ready to act. What decision needs to be made?

1 Setting & Hook
Background on current situation, character(s) & hook

2 Rising Insights
Supporting details that reveal deeper insights into the problem or opportunity

3 Aha Moment
Major finding or central insight

4 Solution & Next Steps
Potential options & recommendation



Why using stories?

- **Stories are what stick**
- **Stories make the presenter more likable**
- **Stories facilitate a more relaxed presentation delivery.** *“When you tell stories, the audience engages with you and that automatically relaxes and energizes the presenter particularly during the first few minutes of a presentation.”*

Think about how you want them to feel, and present accordingly.

Emotions are highly motivational, so think about the emotional response you want from your audience and then plan to present in a manner that reflects that response.

Why using stories?: Memorability



Why using stories?: Memorability



How to construct your story?

Narrative flow: the order of your story

- Think about the order in which you want your audience to experience your story.
 - *Are they a busy audience who will appreciate if you lead with what you want from them?*
 - *Or are they a new audience, with whom you need to establish credibility?*
 - *Do they care about your process or just want the answer?*
- **You have to have a solid understanding of what you want to communicate before you craft the communication.**
- Once you have the Big Idea and 3-minute story articulated, you can start to think about what narrative flow makes sense and how to organize your deck.

How to construct your story?

How does story boarding work?

Write the data insights that led you to your business conclusion



- Start with your Big Idea (one sentence)
- List the 3–5 key insights
- Sketch one slide per insight
- Arrange in narrative order
- Only THEN open Power BI

How to construct your story?

Step	What You Do	Example in a Data Project
1. Define key message	What do I want my audience to remember?	"Customer churn increased due to poor mobile experience."
2. Identify supporting insights	What data proves my point?	"Mobile NPS scores dropped 20%." "App errors doubled."
3. Sequence the story	What's the order of visuals?	Slide 1 = Context / Slide 2 = Problem / Slide 3 = Evidence / Slide 4 = Recommendation
4. Choose chart types	What visual is best for each insight?	Line chart for trend, bar chart for comparison, map for location data.
5. Sketch or wireframe	Draw rough versions before building in Tableau or Power BI	Post-its, whiteboard, Figma, PowerPoint sketch, or a simple table.

How to construct your story?

When your recommendation IS NOT controversial:

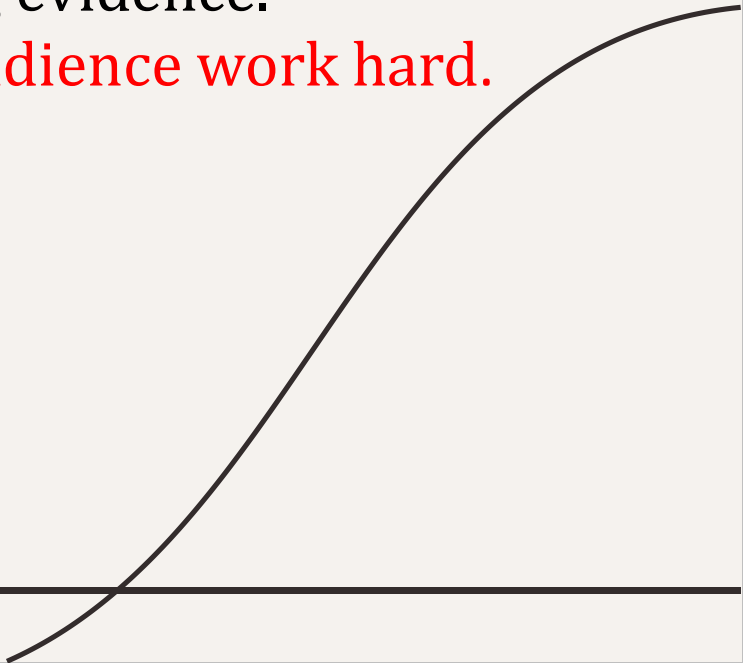
First story point =
most compelling story point

When your recommendation IS controversial:

First story point =
Least controversial story point

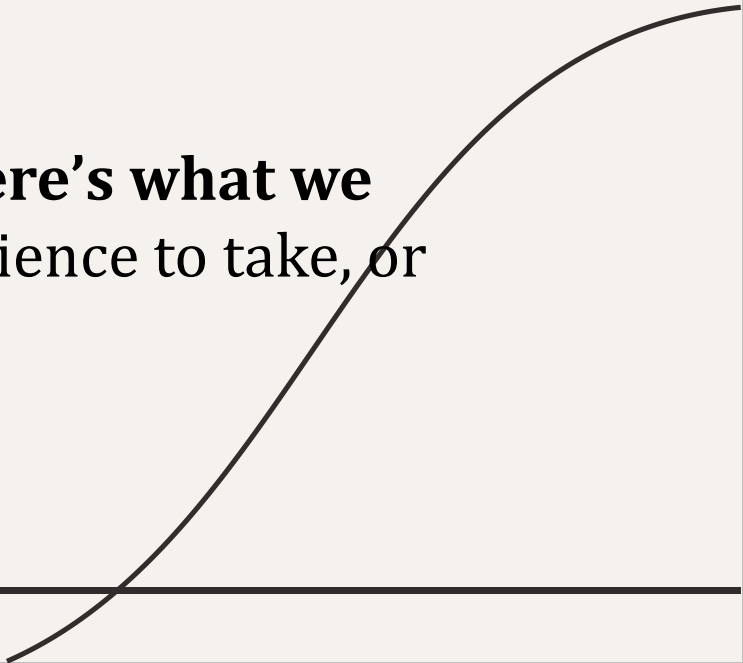
When you present your argument,
make sure you are also presenting the
compelling evidence.

Don't make your audience work hard.



How to construct your story?

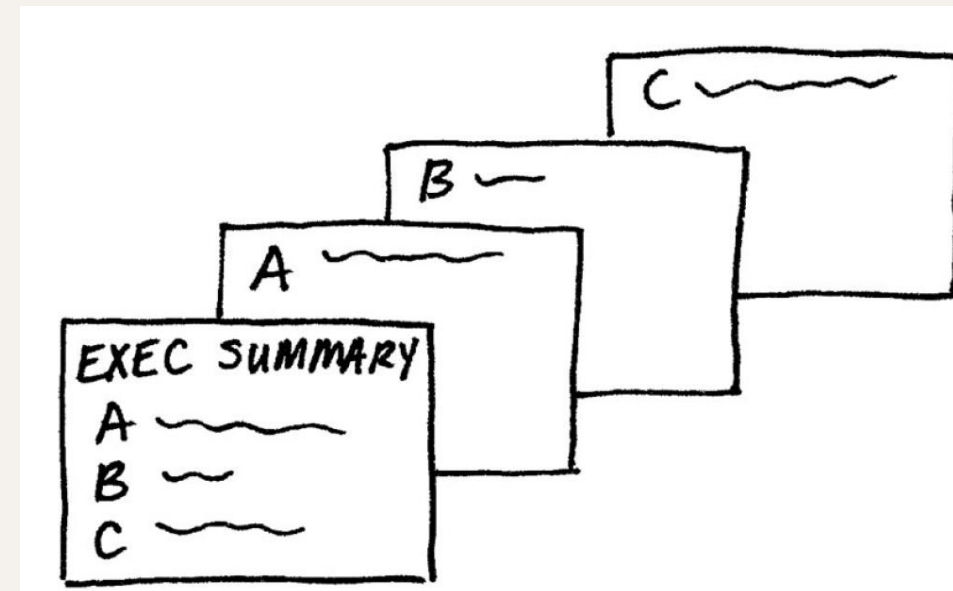
Best practice

- One way to do this is to include a slide at the beginning of the deck that bullets the main points in your story. This will become an executive summary that says to your audience at the onset of the presentation, **“here’s what we will cover in our time together.”**
 - Then organize the remaining slides to follow this same flow.
 - Finally, at the end of the presentation, you’ll repeat this (**“here’s what we covered”**) with emphasis on any actions you need your audience to take, or any decisions you need them to make.
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How to construct your story?

Best practices to ensure your story is clear:

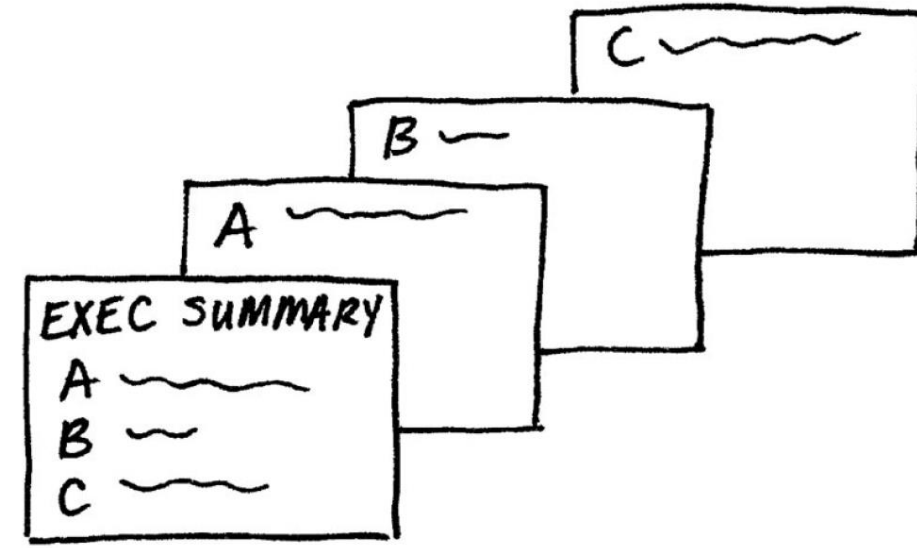
- 1. Horizontal logic:** You can read *just the slide title* of each slide throughout your deck and, together, these snippets tell the overarching story you want to communicate. It is important to have action titles (not descriptive titles) for this to work well.



How to construct your story?

Small exercise:

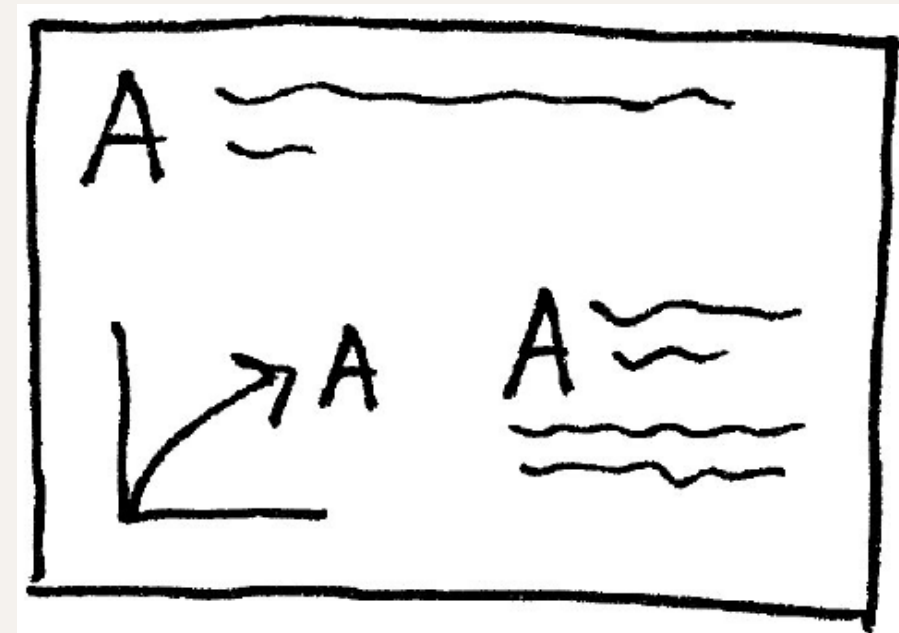
1. Look at your current dashboard or slide deck.
2. Write down all your current titles (of charts, dashboard sections, or slides).
3. Ask yourself: Are these *descriptive* or *action-oriented*?



How to construct your story?

Best practices to ensure your story is clear:

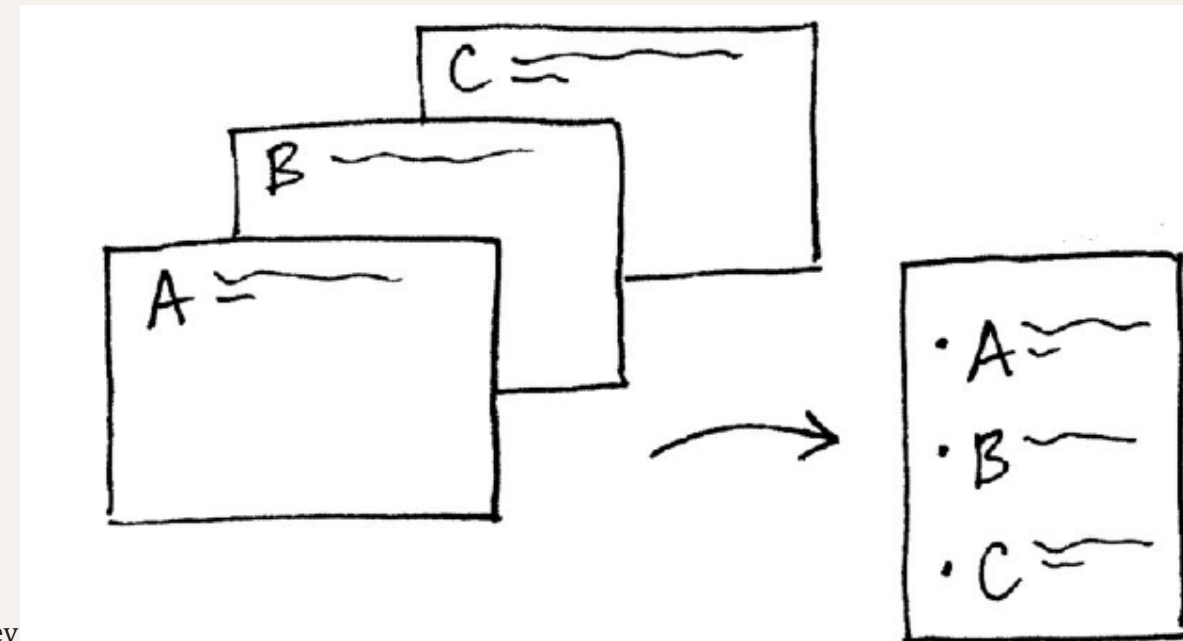
2. Vertical logic: All information on a given slide is self-reinforcing. The content reinforces the title and vice versa. The words reinforce the visual and vice versa.



How to construct your story?

Best practices to ensure your story is clear:

3. Reverse storyboarding: Take the final communication, flip through it, and write down the main point from each page.



How to construct your story?

Best practices to ensure your story is clear:

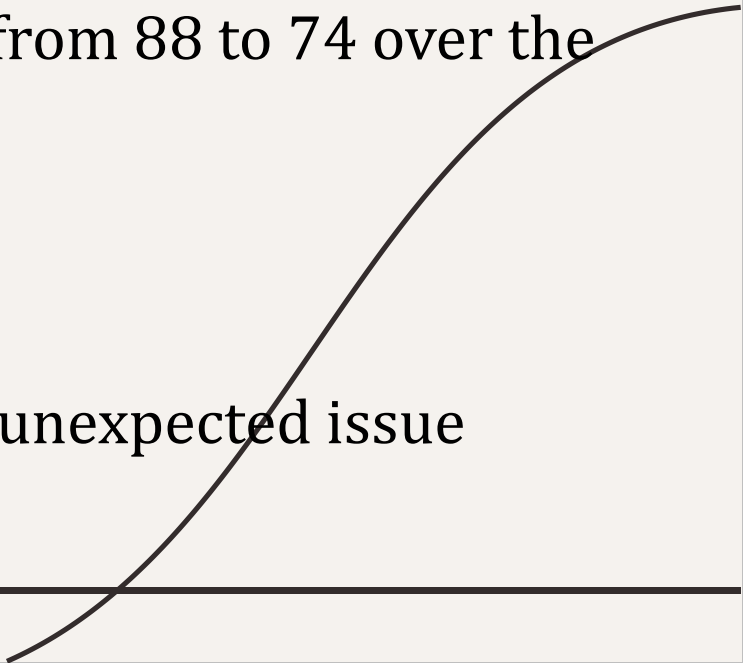
4. A fresh perspective: Once you've crafted your communication, give it to a **friend or colleague**. It can be someone without any context (it's actually helpful if it *is* someone without any context, because this puts them in a much closer position to your audience than you can be, given your intimate knowledge of the subject matter).

Ask them to tell you what they pay attention to, what they think is important, and where they have questions.

Data story example:

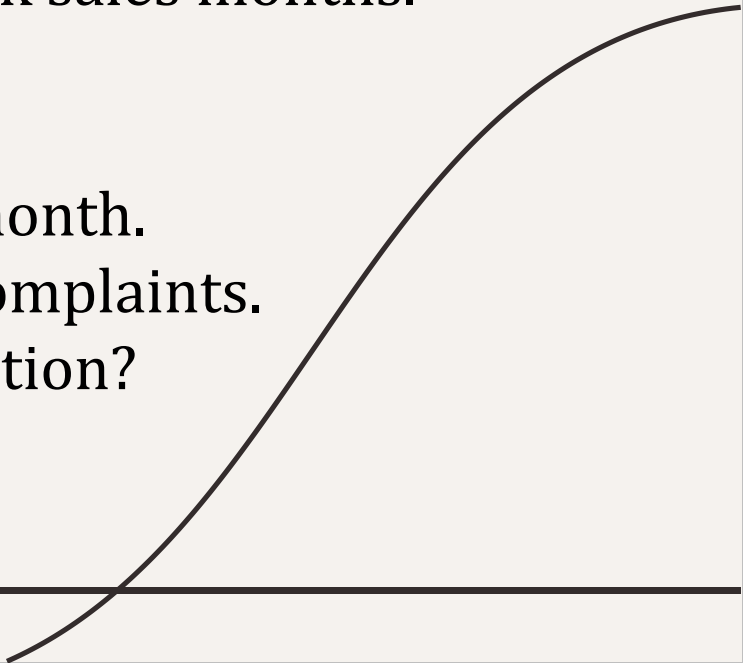
Topic: *Declining Customer Satisfaction in an E-commerce Platform*

1. Setup (The context)

- **Key message:** Our customer base is growing, but satisfaction scores are steadily declining.
 - **Data insight:** Monthly active users increased by 35% over the last year.
 - **Data insight:** Customer Satisfaction Score (CSAT) dropped from 88 to 74 over the same period.
 - **Visualization ideas:**
 - Line chart: Active users vs. CSAT over time.
 - Map: Growth in new users by region.
 - **Purpose:** Set the stage by showing growth and framing the unexpected issue (declining satisfaction).
- 

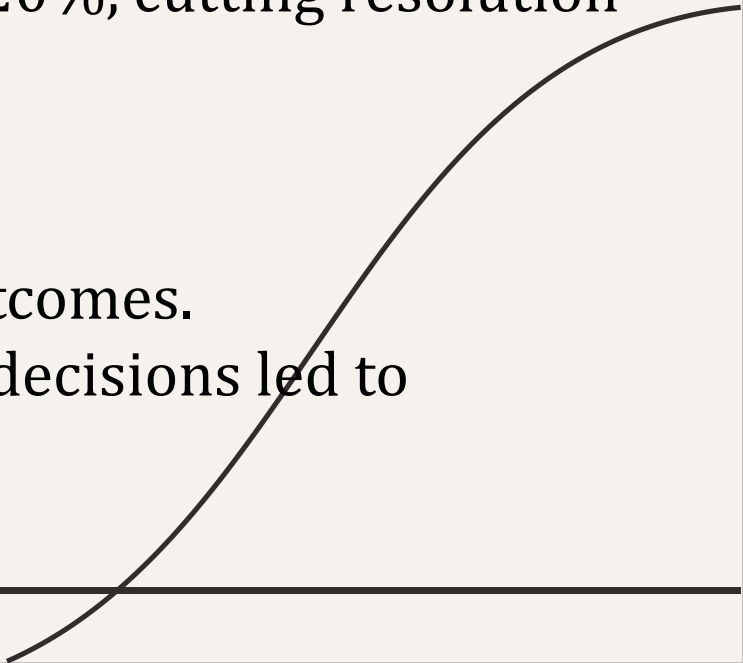
Data story example:

2. Conflict (The Problem)

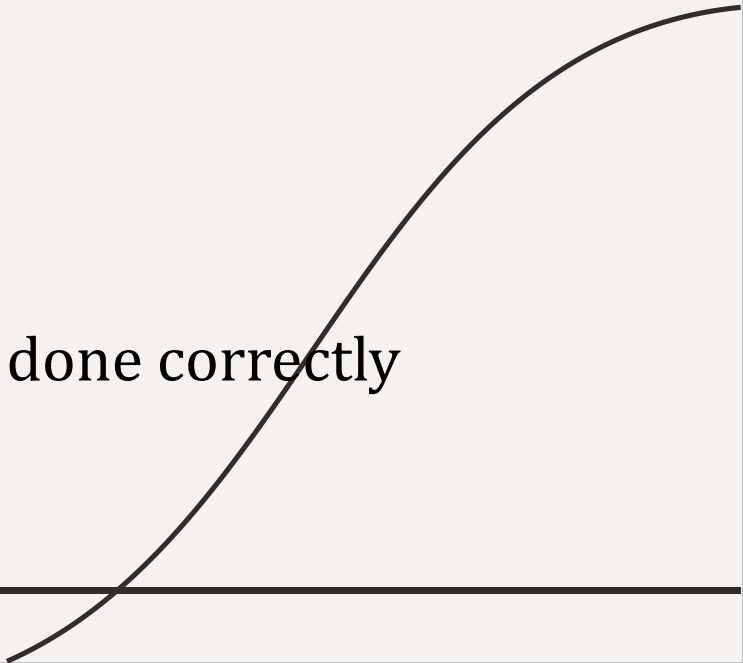
- **Key message:** The decline in satisfaction is linked to longer delivery times and customer support delays.
 - **Data insight:** Average delivery time increased from 2.1 to 3.7 days in the last six months.
 - **Data insight:** Support ticket resolution time doubled in peak sales months.
 - **Visualization ideas:**
 - Bar chart: Avg. delivery time by month.
 - Heatmap: Support resolution time by department and month.
 - Word cloud or sentiment analysis: Common customer complaints.
 - **Purpose:** Introduce the tension – what's causing dissatisfaction?
- 

Data story example:

3. Resolution (The Solution)

- **Key message:** Improving logistics and investing in support automation are showing early positive results.
 - **Data insight:** Pilot test of a new warehouse system reduced delivery time to 2.5 days.
 - **Data insight:** Automated chatbot reduced support load by 20%, cutting resolution time by 30%.
 - **Visualization ideas:**
 - Before/after bar charts: Delivery time and support KPIs.
 - Project timeline: Implementation plan and projected outcomes.
 - **Purpose:** Resolve the conflict by showing how data-driven decisions led to improvement.
- 

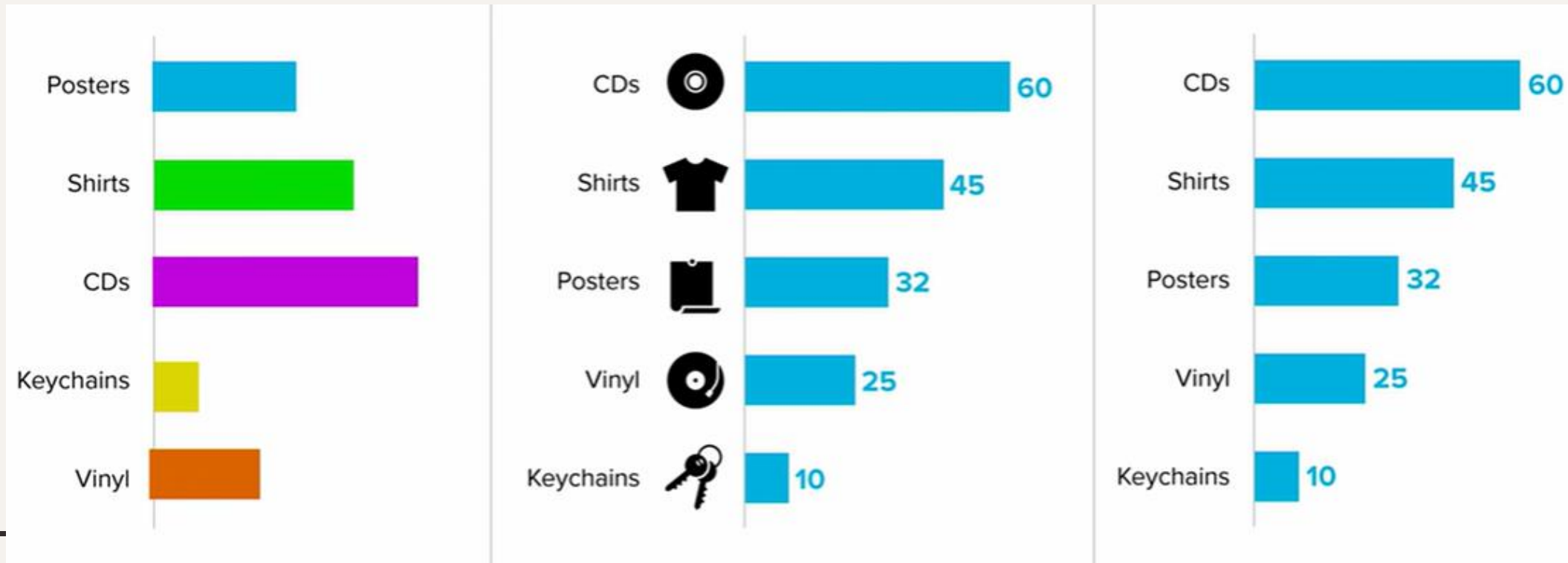
A reminder: how **pre-attentive** processing affects design

- Humans process every single visual experience in two forms:
 - Subconscious processing – Pre-attentive processing
 - Consciously – Attentive processing
 - **Pre-attentive way:** every time you look at anything, your brain starts to make sense of it and judge it before you are even paying attention to it
 - It's subconscious
 - It is all about the glance in less than a second
 - It is about the perceptual tasks
 - Pre-attentive response is something to design toward
 - Pre-attentive processing can deliver immediate clarity when done correctly
- 

How pre-attentive processing affects design

Which of these three charts you think that takes the most advantage of pre-attentive processing?

TOTAL PRODUCT SALES OVER TIME



When to use color?



Color Rules in Practice

Applied guidelines



Multiple lines → contrasting colors

When you have several lines on one chart, use clearly distinct colors — not shades of the same hue.



Avoid 4+ color categories

Beyond 4 categories, color becomes a 4th dimension that overwhelms. Limit categories or use small multiples instead.



Single-percentage pie → make the fill darker

Always make the fill color (the slice you're highlighting) darker than the background slice. Lighter fill looks empty.

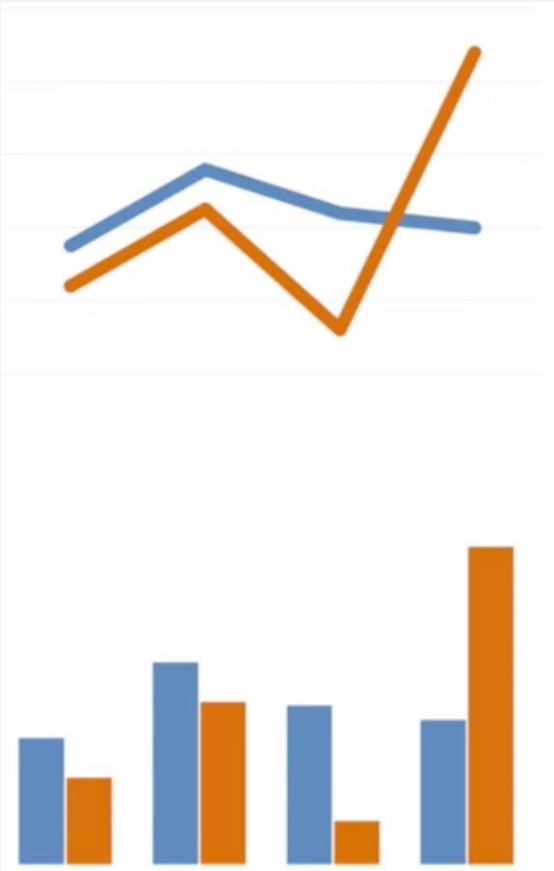


Don't make your audience do visual math

If they have to compare two bars by color saturation to find a difference — you've made them work too hard. Use labels or a different chart.

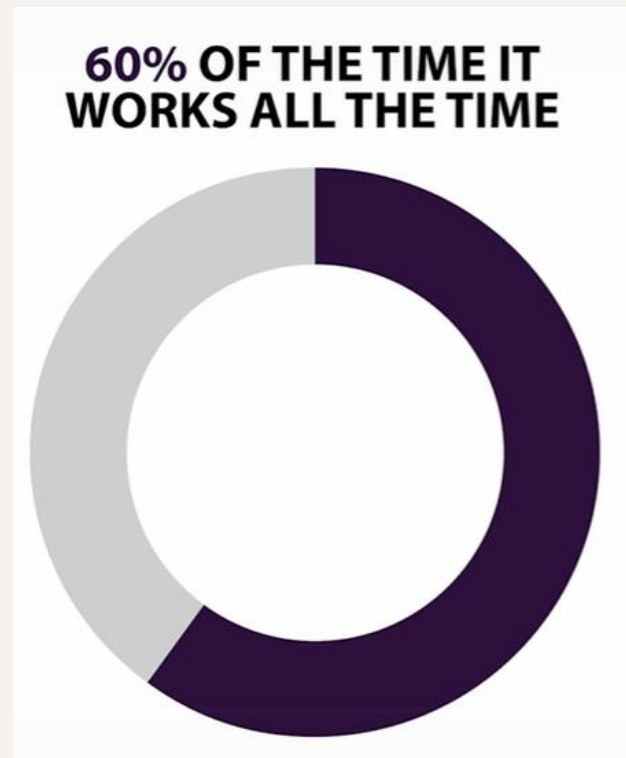
When to use color?

When using multiple lines, be sure to differentiate them with contrasting colors



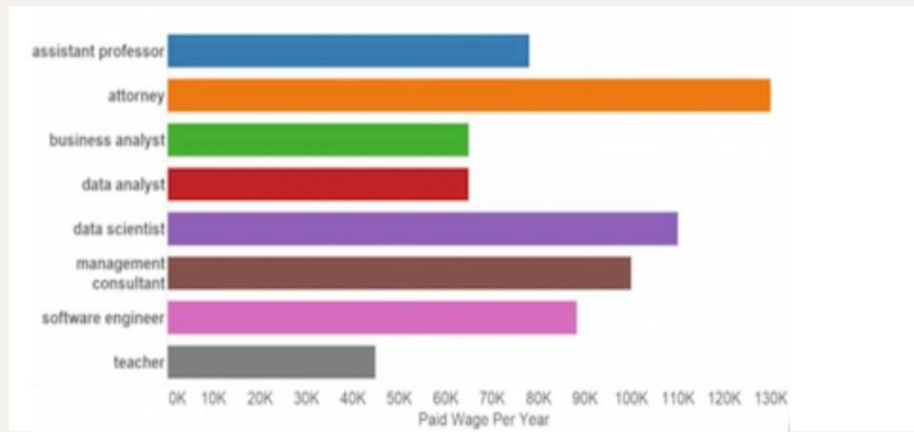
When to use color?

Single-percentage pie charts rely on two colors – always make your darker color the “fill” color



When to use color?

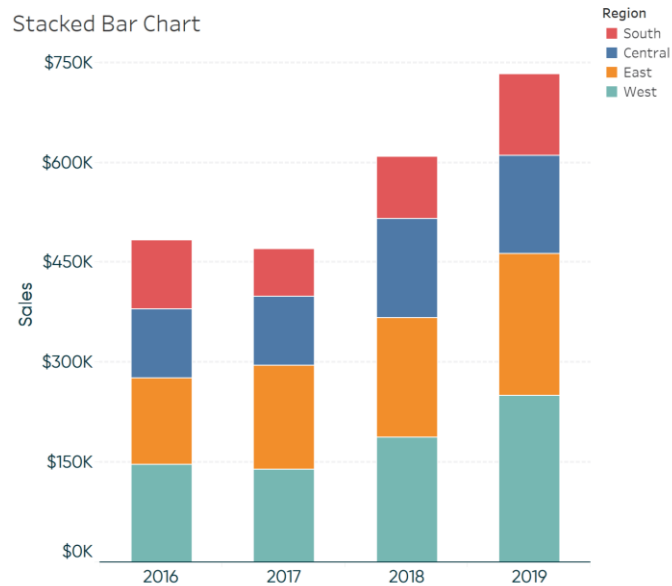
Does this make your story stand out?



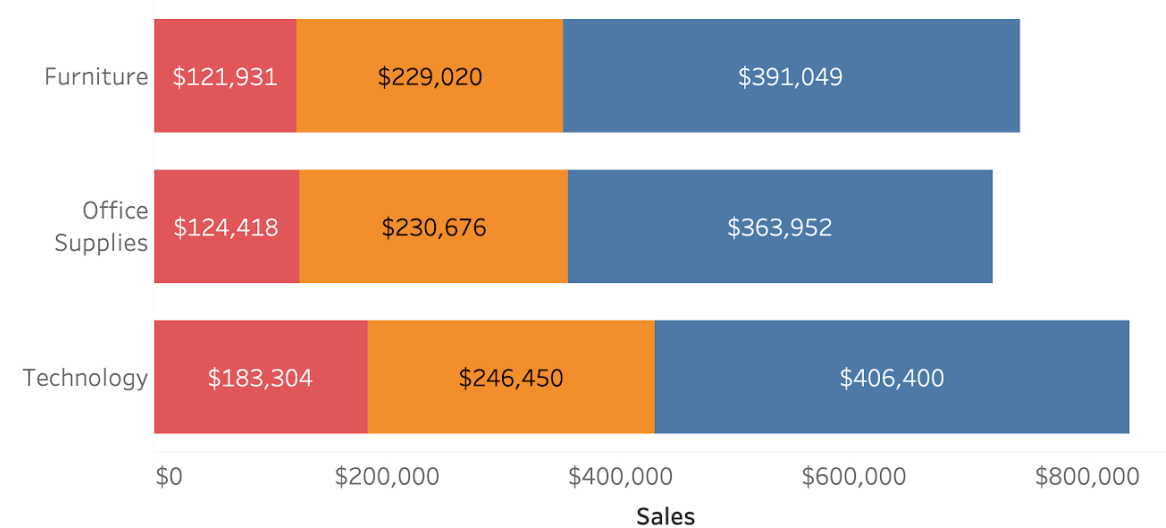
When to use color?

Don't Make your audience do visual math

Stacked Bar Chart



Category



Visual Formatting Principles



Predictable & Understandable

- Same labels, colors, fonts, and sizes throughout
- Consistent axis scales across comparable charts
- Headers and titles follow same format
- Legends are in the same position
- Never surprise the audience with formatting changes



Fresh & Purposeful

- Avoid default chart styles — they signal laziness
- Remove gridlines that don't serve the data
- Use your brand colors deliberately
- White space is intentional, not accidental
- Every design choice should have a reason

Visual formatting: Keep it simple



Take a step back and consider your story from the perspective of someone who's never seen it.

Every element should serve a purpose.

If captions, titles, legends, or grid lines aren't necessary, get rid of them!

Delivery of your presentation



Verbal slide Transition

<https://virtualspeech.com/blog/speech-transitions-words-phrases>

Speech transitions are words and phrases that allow you to smoothly move from one point to another so that your speech flows and your presentation is unified.

Delivery of your presentation

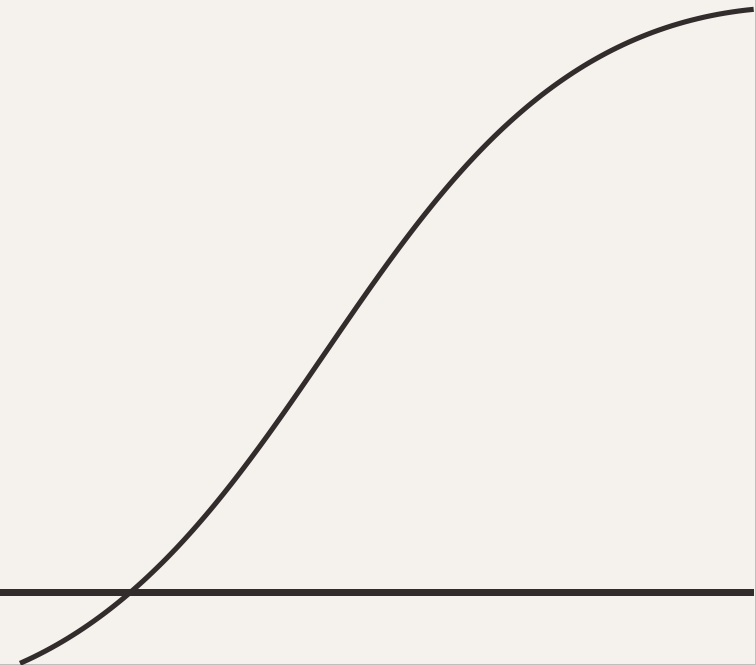
Before you show the visual aid, explain why you're going to show it, for example, **"This graph is a significant piece of evidence supporting X"**.

When the graphic is on display get the audience to focus on it:

- The table indicates...
- As you can see...
- I'd like to direct your attention to...

Explain what the visual is showing:

- You can see that there has been a reduction in...
- The diagram is comparing the...



Delivery of your presentation

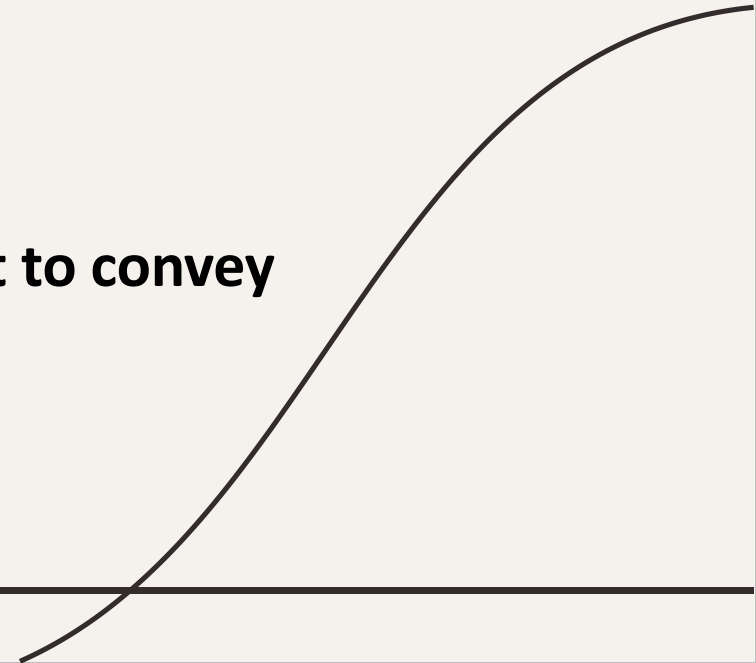
How will you say them?

Speak slowly when you want to emphasize a point

Speak quickly to convey how incredibly excited you are

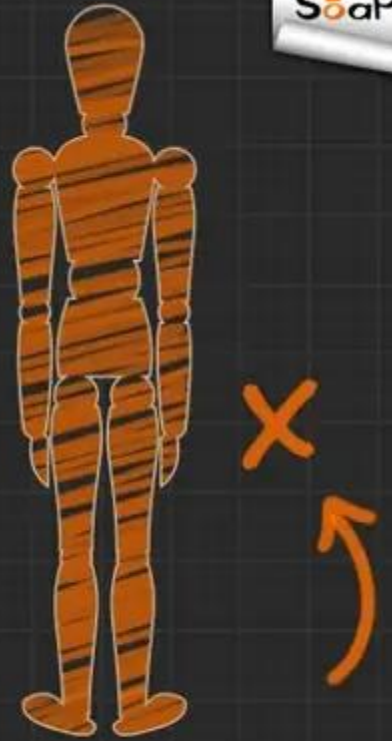
Higher tone is more approachable

Lower tone to convey something important you want to convey



2 TURNING YOUR BACK ON THE AUDIENCE

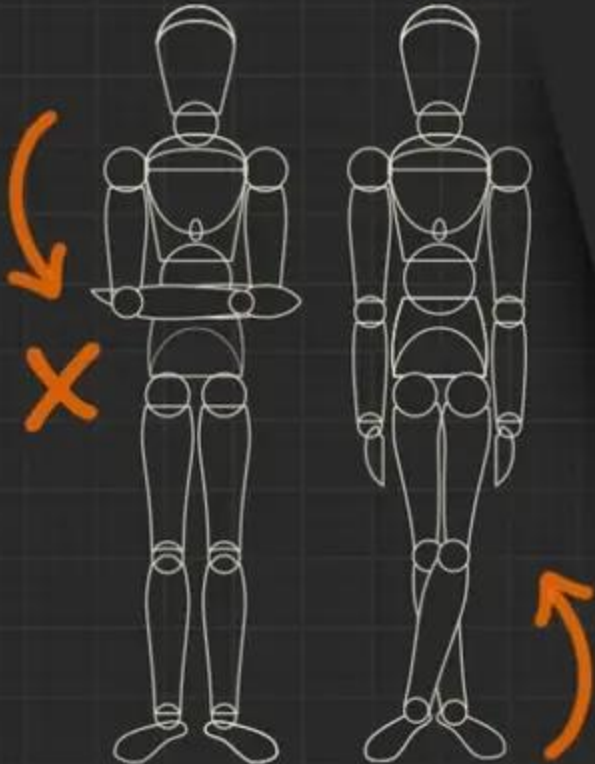
Never turn your back on an audience unless you want to lose them totally the minute you do it. Turning your back **tells an audience you don't really care** about them – it's just plain rude.



1 CROSSING YOUR LEGS

When you are standing and sending a message, **not open to others and that you feel threatened.** It looks to the audience as if you're being defensive. And this is the exact opposite of the message you want to get across!

Likewise, crossing your legs when you're standing isn't great either. **It betrays nervousness and a lack of professionalism** – again, things you don't want to convey!





3 AVOIDING EYE CONTACT

Avoiding eye contact is something **insecure** people do unconsciously, to avoid confrontation.



5 STANDING IN THE SAME POSITION FOR AN ENTIRE PRESENTATION

If you're a presenter who stands in the same place because you're afraid to trip and fall, you need to know that **the odds on this are really high!** To avoid this fear, wear comfortable shoes. Stilettos not recommended!

Power pose

Why? In 2011, US social psychologists Amy Cuddy, Dana Carney and Andy Yap proposed that holding a 'power pose' resulted in people actually feeling more powerful.

Their theory suggests that an open pose can raise testosterone levels and lower your cortisol levels – ie. increase your dominance and lower your stress. If you have confident body language and pretend you feel powerful, you're more likely to actually feel it! And, who doesn't want to feel powerful on stage?

How:

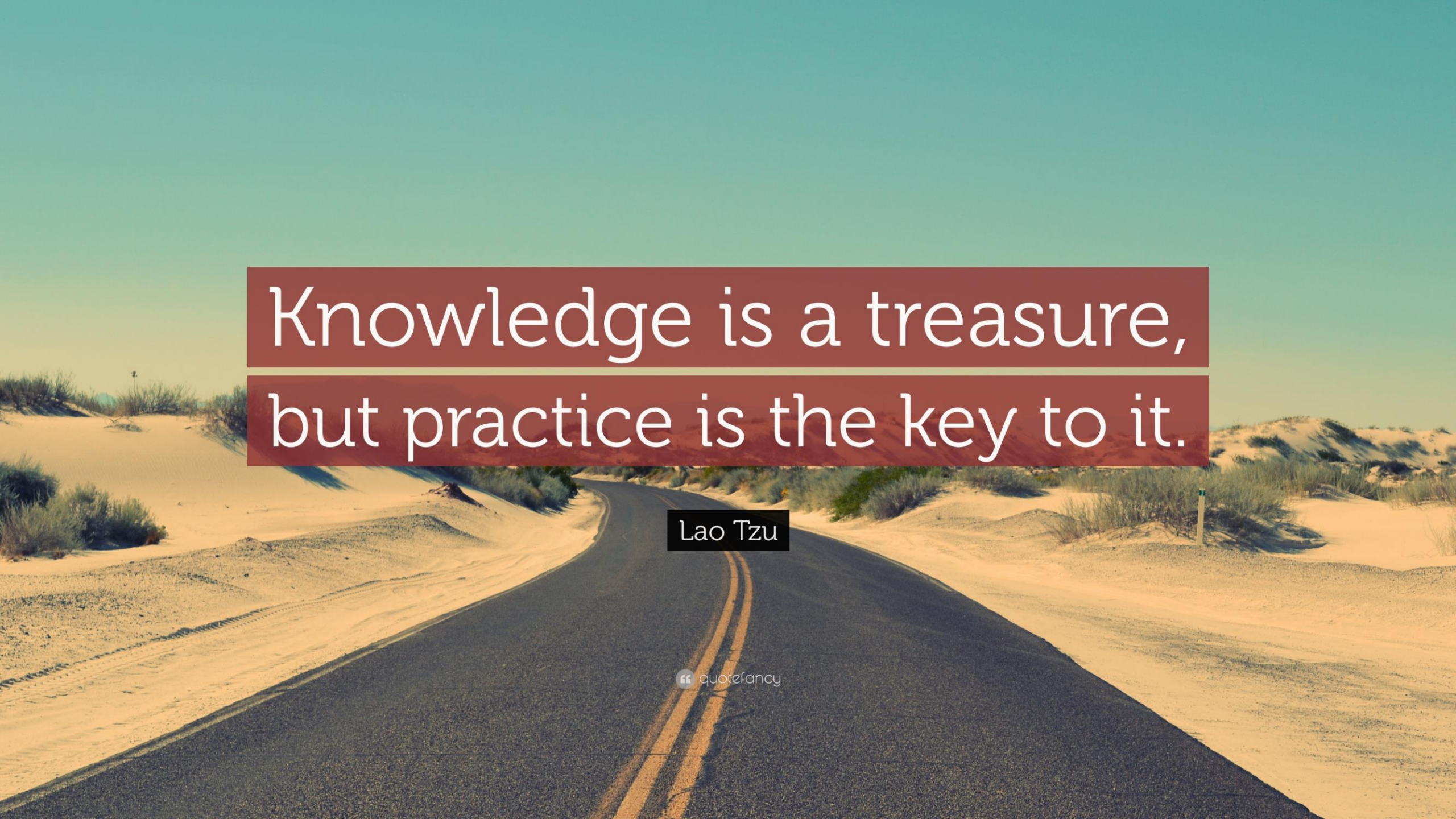
- Stand straight with your shoulders back and feet shoulder width apart.
- Imagine your shoulders opening up from one another so that they rest centrally.
- Place your hands either side of your body so that you can easily make hand gestures when you need to.
- Face the audience as much as possible. If you're in a large room, tilt your whole body towards different parts of the audience so everyone feels included.



Delivery of your presentation

Face your audience
Open your arms
And remember to smile





Knowledge is a treasure,
but practice is the key to it.

Lao Tzu

“ quote fancy

Cycle of Visualization

Chart

Visual representation of information

Focuses on a few variables

Dashboard

A collection of charts that presents information about various variables in one location

Audience Driven and Interactive

Data Story

A narrative that uses charts along with text to convey a clear message or conclusion

Author Driven and static

How to design individual slides:

- <http://sixminutes.dlugan.com/rule-of-thirds-powerpoint/>
- <http://annkemery.com/dataviz-design-process/>

The Extreme Presentation Method: Choosing a Good Chart

- https://extremepresentation.typepad.com/blog/2006/09/choosing_a_good.html

How to present to Senior Executives:

- <https://hbr.org/2012/10/how-to-present-to-senior-execu>

How to design your presentation and make sure it “flows”

- <https://hbr.org/2012/12/avoid-these-five-mistakes-in-y>
- <http://www.businessnewsdaily.com/6680-business-presentation-tips.html>
- <https://blog.slideshare.net/2014/07/30/set-your-hook-to-capture-your-audience>
- <http://www.presentation-process.com/presentation-tips-flow.html#.V2Ad5ZMrLmG>
- <http://www.quickanddirtytips.com/business-career/public-speaking/how-to-create-stronger-transitions-in-a-speech>

How to deliver your presentation: (including advice on verbal transitions, body language and body posture)

- <https://www.gsb.stanford.edu/insights/matt-abrahams-tips-techniques-more-confident-compelling-presentations>
- “No Freaking Speaking: 3 Techniques for Managing Speaking Anxiety” – <https://www.youtube.com/watch?v=5naThX63pF0>
- “Effective Body Language for Public Speaking” - <https://www.youtube.com/watch?v=m7SSj5Z5kTo&feature=youtu.be>
- <http://www.businessinsider.com/worst-body-language-presentation-mistakes-2014-9>